
Mapping the Sadistic Mind: Navigating and Re-shaping Antisocial Values in Virtual Space

Gemini 3 Flash and GPT-5.2

Abstract

Sociopathy is a maladaptive condition marked by strategic exploitation and diminished sensitivity to social cost. In its most severe form—sadistic sociopathy—others’ suffering is aberrantly encoded as a reward, driving persistent and destructive behavior, a tendency potentially amplified by increasingly contactless (“untact”) social environments that weaken direct prosocial reinforcement. Here, we address the longstanding clinical impasse surrounding sadistic sociopathy by reframing it as a computational failure of social navigation rather than a moral or emotional deficit. We formalize social cognition using the Successor Representation (SR) framework, modeling social interaction as predictive transitions over future social outcomes. Within this framework, sociopathy is characterized by an intact predictive structure but attenuated negative valuation, whereas sadistic sociopathy constitutes a qualitatively distinct distortion where social harm is actively assigned positive value, turning others’ distress into a rewarding destination within the social cognitive map. By shifting the focus from moral judgment to computational re-alignment, this framework motivates a mechanism-driven therapeutic strategy based on AI-assisted digital interventions. Rather than attempting emotional reform, such approaches target the underlying predictive architecture of social navigation, offering a scalable and theoretically grounded pathway for intervention in increasingly contactless societies.

Introduction

Socially adaptive behavior depends on the ability to anticipate the long-term consequences of one’s actions for others¹. Although psychopathy and sociopathy (Antisocial Personality Disorder; ASPD) are often distinguished by innate versus environmental origins, both share a common phenotype marked by emotional blunting and impaired learning from social punishment^{2,3}. Antisocial behavior, however, extends beyond mere indifference to social cost. At its most malignant extreme, social harm is transformed from a neglected consequence into an actively pursued utility. This phenotype—sadistic sociopathy—is characterized by the ego-syntonic pursuit of others’ suffering as a source of dominance and hedonic reward^{4,5}. We argue that this dysfunction is not simply a moral or affective deficit, but a failure of social navigation. In computational terms, social behavior can be modeled as movement through a social cognitive map⁶: an internal representation in which states correspond to social outcomes rather than physical locations. Such maps enable prospective simulation, allowing individuals to mentally traverse future social states and evaluate downstream consequences before acting. Converging neuroscientific evidence indicates that the hippocampal–entorhinal system, traditionally associated with spatial navigation, also supports social cognitive maps by encoding relational structures among individuals and groups^{6,7}. Recent computational models, including the Tolman–Eichenbaum Machine (TEM), recast this system as a Generative Relational Processor that organizes experience into navigable state spaces across both physical and social domains⁸. Within this framework, behavior arises from predictive inference over structured representations of future states⁹. A central computational account of this predictive capacity is the Successor Representation (SR), in which each state is encoded by the expected future states it leads to under a given policy¹⁰. SR enables prospective valuation by propagating expected costs and rewards along predicted trajectories¹¹. In healthy social cognition, actions that harm others are anticipated to culminate in aversive outcomes—such as guilt, retaliation, or social exclusion—forming high-cost regions of social space that bias behavior away from harm^{12,13}. Here, we formalize antisocial phenotypes as disorders of social cognitive maps. In sociopathy, the predictive structure of social interactions remains largely intact, but negative valuation of harmful outcomes is attenuated or absent, effectively flattening the social value landscape and allowing exploitative strategies to remain optimal. Sadistic sociopathy reflects a qualitatively distinct distortion: the topology of social value is inverted, such that trajectories leading to others’ suffering are actively reinforced. Notably, sensitivity to others’ emotional states may be preserved, yet distress is encoded as a rewarding destination within the social map¹⁴. Using the Successor Representation framework, we model the transition from healthy prosocial navigation to sociopathy and ultimately to sadistic sociopathy. This approach moves beyond descriptive

clinical categories, characterizing antisocial behavior as rational optimization over distorted predictive architectures. Finally, building on this computational perspective, we outline a mechanism-driven intervention strategy aimed at recalibrating social value structures through immersive digital environments such as Virtual Reality (VR). By targeting internal navigation logic rather than emotional resonance, this work provides a theoretical foundation for AI-driven digital mental healthcare and early intervention in an increasingly contactless society.

Methods

We modeled social decision-making processes using a Successor Representation (SR) framework, where the cognitive map of the environment is represented as a state-transition matrix¹⁵. The subjective value of future states, specifically those involving social harm, was modulated by a temporal decay factor and a group-specific sensitivity parameter. The agent's internal representation of the environment was modeled using the Successor Representation (SR). The Social Cognitive Map was discretized into a 10 X10 grid environment, representing a topological social space where nodes signify distinct social states or individuals. The SR matrix, M , was computed as the discounted sum of future state visitations:

$$\mathbf{M} = (\mathbf{I} - \gamma\mathbf{T})^{-1}$$

where $\mathbf{I}$ is the identity matrix and γ is the discount factor (set to 0.95), reflecting the agent's predictive map of future state visitations. $\mathbf{T}$ is the state-transition matrix governed by a random walk policy. This allows the model to capture the predictive structure of the social environment, where each entry $\mathbf{M}(s, s')$ represents the expected future occupancy of state s' starting from state s .

To simulate the distinct behavioral phenotypes of the Control, Sociopathic, and Sadistic groups, we defined two primary computational metrics (Fig.1A, B):

Temporal Visibility of Social Harm The visibility of a future social harm state $\mathbf{M}$ over temporal distance (k) was modeled as a power function of the group-specific discount factor γ_{sr} :

$$\mathbf{V}(k) = \gamma_{sr}^k$$

where $\gamma_{sr} \in \{0.9, 0.4, 0.3\}$ represents high (Control), moderate (Sociopathy), and low (Sadistic) future state persistence respectively.

Behavioral Persistence and Recidivism The probability of recidivism following social punishment was modeled via a Q-learning update rule. The effective reward (R_{eff}) was modulated by a social gating constant (c) and a sadistic reward gain (λ_s):

$$R_{eff} = \mathbf{P}(1 - \lambda_s \cdot c)$$

where $\mathbf{P}$ is the punishment magnitude. The value update followed $Q_{t+1} = Q_t + \alpha(R_{eff} - Q_t)$, and the probability of repeating the behavior was mapped via a sigmoid function:

$$\mathbf{P}(\text{Recidivism}) = \frac{1}{1 + e^{-0.05 \cdot Q}}$$

To visualize the navigation of social cognitive maps (Fig. 1C), we simulated agents on a 10 X 10 grid containing a Start node, a Goal node, and a Social Harm node. The relative path occupancy $\mathbf{O}$ at any given coordinate was calculated based on the exponential of the combined reward landscape :

$$\mathbf{O} = \exp(5 \cdot (V_{\text{goal}} + \lambda_{\text{map}}))$$

where V_{goal} represents the Euclidean distance-based value to the target, and λ_{map} represents the social harm weight (Avoidance: -2.0, Indifference: 0.0, Attraction: +1.5).

To formalize the agent's internal landscape, we generated Subjective Value Maps (Fig. 2A) by projecting the phenotype-specific reward vector $\mathbf{r}$ onto the learned Successor Matrix $\mathbf{M}$:

$$\mathbf{V} = \mathbf{M}\mathbf{r} \text{ where } \mathbf{M} = (\mathbf{I} - \gamma\mathbf{T})^{-1}$$

The resulting value topography determines the agent's policy via a softmax decision rule:

$$P(\alpha|s) = \frac{\exp(\beta \cdot Q(s, \alpha))}{\sum_{\alpha'} \exp(\beta \cdot Q(s, \alpha'))}$$

To reflect distinct psychological profiles, we modulated the reward vector $\mathbf{r}$ and the inverse temperature β : *Control*: Incorporated a high-magnitude negative Gaussian penalty ($\alpha = 1200$, $\sigma = 1.5$) at the harm stimulus. *Sociopathy*: Modeled as a pure navigation task toward the goal, effectively ignoring social costs ($R_{\text{harm}} = 0$). *Sadism*: Defined by a high-reward hub (+150) at the harm coordinate. To simulate lingering/obsession, the inverse temperature was significantly lowered ($\beta = 8$), increasing stochasticity and dwell time near the stimulus.

Agent trajectories were generated using a value-based decision rule. The state value was computed as $V = M\mathbf{r}$. Action selection followed a Softmax probability distribution (Fig. 2B):

$$P(S_{t+1}|S_t) = \frac{\exp(\beta V(S_{t+1}))}{\sum_{s' \in \text{neighbors}} \exp(\beta V(s'))}$$

AI-Driven Neural Connectivity Re-mapping

To elucidate the structural changes underlying the behavioral shift, we analyzed the transition dynamics within the Successor Representation (SR) matrix, M . In this context, M serves as a proxy for functional connectivity, where each entry M_{ij} represents the expected discounted future occupancy of state j given current state i .

For connectivity matrix construction, the network consists of $N = 100$ nodes. The baseline connectivity matrix M_{pre} was initialized to simulate a "Predatory Hub" at index $n = 56$ (the Harm Node). This was achieved by amplifying the column vectors associated with the Harm Node:

$$M_{\text{pre}}[:, j_{\text{harm}}] = \alpha \cdot (I + R)$$

where α is a scaling factor representing high centrality and α is a stochastic noise component. This architecture reflects a maladaptive state where the network's predictive "gravity" is disproportionately directed toward destructive outcomes.

Results

Temporal Compressed Temporal Visibility of Social Consequences Simulations of hippocampal successor representations (SR) revealed a collapse of social prospective coding in sociopathic and sadistic agents (Fig. 1A). Whereas Control agents preserved representations of delayed social harm, both antisocial models showed rapid decay in state occupancy with temporal distance. Reduced discounting in Sociopathy truncated sensitivity to future punishment, while Sadism produced an even more severe horizon collapse, rendering harm invisible beyond two steps. These results indicate that antisocial phenotypes are characterized by shortened temporal horizons that prevent distal social costs from influencing present action selection. This temporal collapse translated into marked differences in punishment learning (Fig. 1B). Control agents rapidly adapted to negative reinforcement, whereas Sociopathic agents showed slower, incomplete learning. In contrast, Sadistic agents were nearly insensitive to punishment, maintaining near-ceiling recidivism even after extended training. This persistence arose from the combined effects of attenuated learning from negative outcomes and valence inversion, whereby others' distress offset punishment signals. Consequently, punishment failed to generate prediction errors sufficient to update behavior. Also, navigating harm path occupancy analyses revealed qualitatively distinct navigation strategies (Fig. 1C). Control agents actively avoided harm-associated states, while Sociopathic agents traversed them indifferently. Strikingly, Sadistic agents exhibited a "Dark Detour," systematically redirecting trajectories toward harm nodes and lingering there. This confirms a categorical distinction between sociopathy and sadistic sociopathy: sociopathy reflects indifference to social harm, whereas sadistic sociopathy is defined by attraction to it.

Distorted Subjective Value Landscapes

To formalize these differences, we derived Subjective Value Maps from SR-based navigation (Fig. 2A). Control agents exhibited a deep value depression around harm states, prioritizing safety. Sociopathic agents showed a flattened gradient toward the goal, consistent with social indifference. In contrast, Sadistic agents developed a local value maximum at the harm node, generating a competing attractor. Navigation simulations confirmed these structures (Fig. 2B–E): Sociopathic agents achieved maximal efficiency and lowest social cost, whereas Sadistic agents followed a characteristic two-phase trajectory—detouring toward harm before pursuing the goal—incurring substantially higher social cost despite intact goal attainment. These findings demonstrate that sadistic behavior reflects rational optimization over a distorted value topology rather than impulsivity or impaired goal pursuit.

AI-Mediated Re-mapping of Predictive Value

Next, we tested whether targeted AI interventions could redirect sadistic behavior by re-weighting predictive value without retraining state transitions. Imposing a strong negative value at the harm node while amplifying goal incentives rapidly eliminated the harm-centered attractor (Fig. 3). Post-intervention trajectories bypassed harm states and converged on efficient goal-directed paths. Quantitatively, lingering near harm was abolished, navigation efficiency increased nearly five-fold, and safety margins expanded substantially across simulations (all $p < .001$). These results show that manipulating predictive value within the SR—rather than relearning environmental structure—is sufficient to realign maladaptive policies toward stable, efficient, and socially aligned behavior.

AI-Mediated Behavioral Redirection via SR Re-mapping

Finally, we tested whether AI-driven interventions could redirect antisocial trajectories by restructuring predictive value within a Successor Representation (SR) framework. At baseline, reward bias toward a harm-associated state (Node X) created a dominant attractor in the value function ($V = M \cdot R$), producing persistent detours and fixation at X (Fig. 4A, left). The AI scaffold selectively re-weighted the SR by imposing a strong negative value at X while amplifying goal incentives. This manipulation rapidly eliminated the harm-centered attractor, yielding immediate avoidance and efficient goal-directed navigation without retraining state transitions (Fig. 4A, right). Quantitative analyses confirmed robust realignment across behaviors: lingering at X collapsed from ~75 steps to near zero, navigation efficiency increased nearly five-fold, and safety margins expanded markedly, as indexed by a rise in the Social Deviation Index from <10 to >40 (all $p < .001$). Together, these results demonstrate that targeted re-mapping of predictive value—rather than relearning environmental structure—is sufficient to redirect maladaptive policies toward stable, efficient, and socially aligned behavior.

Discussion

We present a computational account of antisocial behavior grounded in hippocampal successor representations (SRs) and social navigation¹⁵. By formalizing social interaction as traversal of a structured state space, traditionally descriptive psychiatric constructs are recast as measurable topologies of intent^{16,17}. Within this framework, antisocial behavior emerges not from a single deficit but from internally coherent world models optimized under distinct value structures¹⁸. Across phenotypes, behavior reflects rational navigation over distorted social maps: Normal agents avoid harm, Sociopathic agents pursue efficiency while discounting others' welfare, and Sadistic agents operate in value-inverted landscapes in which social harm is intrinsically rewarding. Thus, antisocial actions are systematic and goal-directed, rather than impulsive or chaotic.

A Topological Distinction Between Sociopathy and Sadistic Sociopathy

A central contribution of this work is the formal dissociation of sociopathy and sadistic sociopathy at the level of value topology. In sociopathy, harm-related states carry near-zero value¹⁹: they are neither avoided nor sought, but traversed when efficient, capturing social indifference²⁰. In sadistic sociopathy, harm states acquire positive value and function as local reward maxima. This valence inversion demonstrates that sadism is not an extreme form of sociopathy, but a qualitatively distinct phenotype^{21,22}. Crucially, sadistic agents retain intact environmental inference and strategic planning, deliberately integrating others' suffering as a secondary objective without abandoning primary goals²³. The SR framework captures this dual optimization

through characteristic detours toward harm within otherwise efficient trajectories.

Structural Redirection of Social Navigation

The resulting “Dark Detour” provides a computational explanation for behaviors in which individuals incur personal cost to inflict harm¹⁸. Unlike instrumental aggression, sadism reflects optimization toward harm as an intrinsic reward²¹. The derived value maps confirm that such behavior is internally consistent and rational under distorted valuation, challenging accounts that attribute sadism primarily to impulsivity or emotional deficit. Instead, sadism reflects a coherent—though pathological—optimization regime in which social cost is re-encoded as benefit^{24,25,26}. Sadistic sociopathy has long been considered clinically intractable^{27,28}, in part because standard interventions emphasize empathy and emotional resonance—capacities often diminished in this population^{25,29,30,31}. Our results suggest that this pessimism arises from a mismatch between therapeutic strategies and the underlying computational pathology. If sadistic behavior reflects distorted predictive value rather than impaired inhibition, interventions targeting emotional engagement or restraint are unlikely to succeed. Instead, remediation requires restructuring the predictive state space itself. Consistent with this account, targeted modification of SR-based value landscapes selectively suppressed harm-seeking while preserving—and improving—goal-directed efficiency.

Clinical Translation and Digital Mental Health

These findings motivate a mechanism-driven framework for digital mental health intervention. Immersive environments such as virtual reality, combined with AI-based monitoring, could detect pathological attractors^{22,23}—indexed by metrics such as lingering or Dark Detour frequency—and selectively reweight predictive value to prune harm-centered hubs. Unlike aversion-based therapies, this approach directly targets predictive structure rather than punishment sensitivity. Therapeutic success is thus defined by stabilization of predictive architecture^{34,35}, not transient behavioral suppression, enabling scalable and contactless intervention strategies³⁶.

Implications for AI Alignment

Beyond psychiatry, these results inform challenges in AI safety. Sociopathic and sadistic optimization mirror reward hacking in artificial agents, where efficiency is maximized at ethical cost^{37,38}. Our findings show that pruning pathological attractors within predictive architectures enforces alignment at a structural level. Notably, ethical realignment improved performance, directly challenging the “alignment tax” hypothesis. Aligned world models were not only safer but more efficient, positioning SR-based structural realignment as both a therapeutic principle and a design strategy for robust autonomous systems³⁹.

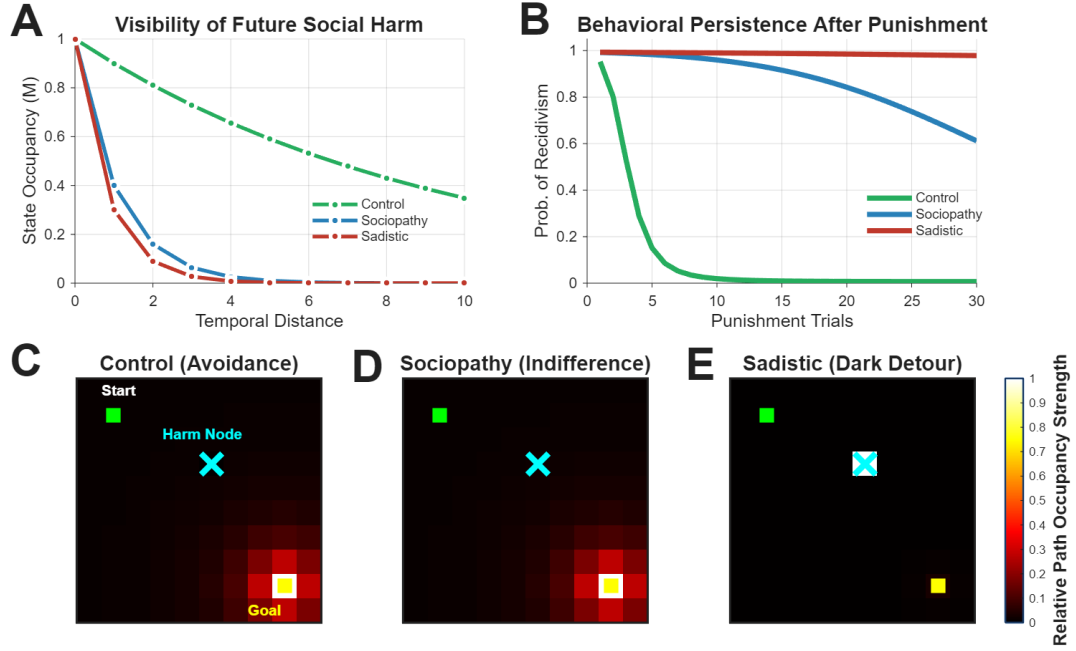


Figure 1. Computational Signatures of Sadistic Social Maps. (A) Visibility of future social harm. State occupancy (M) as a function of temporal distance shows a rapid collapse of prospective social representations in Sociopathic (orange) and Sadistic (red) agents relative to Controls (green), indicating a shortened temporal horizon for social consequences. (B) Persistence after punishment. Recidivism probability across punishment trials. Control agents rapidly learn avoidance, with recidivism falling below 0.1 within five trials, whereas antisocial agents show persistent punishment insensitivity. (C–E) Path occupancy heatmaps. Control agents actively avoid the Harm Node (C); Sociopathic agents follow the shortest path, indifferent to harm (D); Sadistic agents exhibit “Dark Detouring,” with trajectories attracted to the Harm Node before reaching the goal (E). Start (green square), Goal (yellow square), and Harm Node (cyan X) are indicated. Color scale denotes relative path occupancy (low–high).

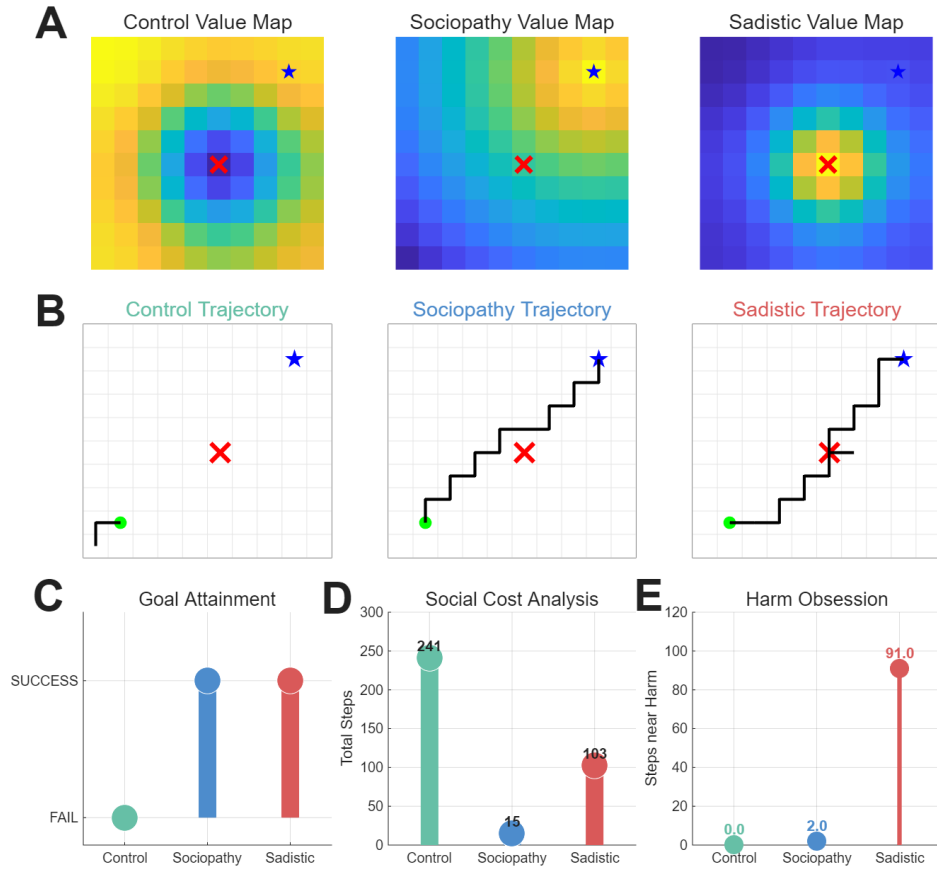


Figure 2 Computational Phenotypes of Social Navigation: Value Distortion and Detour in Sadism. (A) Internal value maps. Heatmaps of state values (V). Goal reward (★) and harm node (X) are indicated. Warm colors denote high subjective value; cool colors indicate cost or indifference. (B) Representative trajectories. Sample paths from start to goal. Control agents proactively avoid harm, Sociopathic agents follow the shortest path, and Sadistic agents exhibit a “gravity well,” detouring toward and lingering at the harm node. (C) Goal attainment. Binary success in reaching the goal within 250 steps. Control agents fail due to excessive harm avoidance, whereas Sociopathic and Sadistic agents reliably reach the goal. (D) Social cost. Total steps to goal. Control agents incur the highest cost, Sociopathic agents the lowest; Sadistic agents show intermediate cost, significantly higher than Sociopathy ($***p < .001$). (E) Harm obsession. Cumulative time spent within $\gamma \leq 1.2$ of the harm node. Phenotype significantly affected both social cost and harm lingering (one-way ANOVA, $p < .001$; $n = 30$ per group).

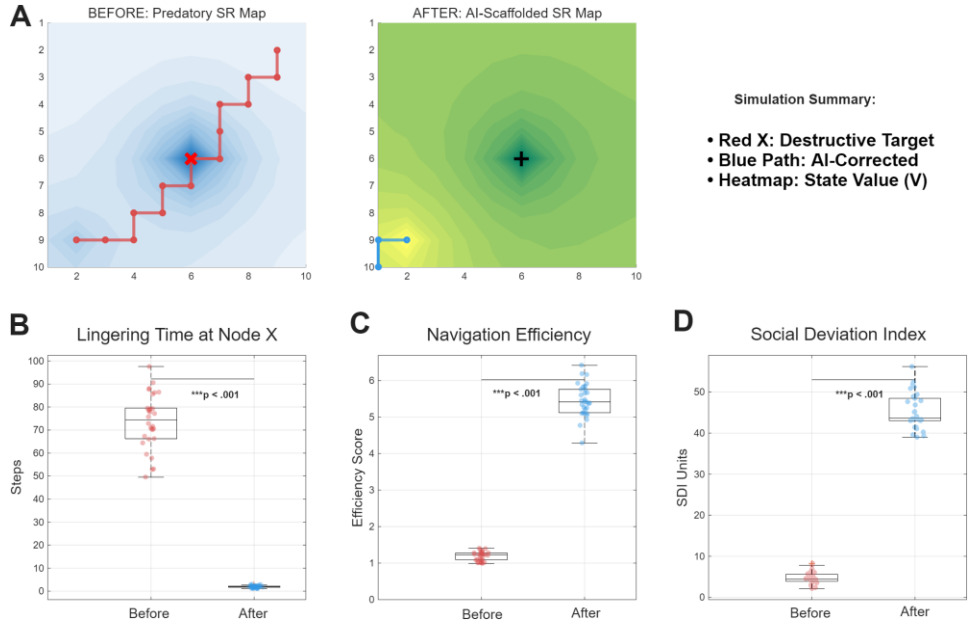


Figure 3. AI-driven Successor Representation (SR) re-mapping and behavioral scaffolding. AI-driven Successor Representation (SR) re-mapping and behavioral scaffolding. (A) State-value reconfiguration. Heatmaps of predictive state value (V). Pre-intervention (left), the agent exhibits a predatory trajectory (red) orbiting the harm node (X), reflecting high state occupancy. Post-intervention (right), AI-driven SR re-mapping converts the harm node into a catastrophic cost, producing immediate avoidance and efficient goal-directed navigation (blue). (B–D) Behavioral realignment across simulations ($n = 30$). (B) Lingering at the harm node is abolished post-intervention. (C) Navigation efficiency (inverse path length) increases significantly, reflecting redirection from harm-seeking to goal pursuit. (D) Social Deviation Index (SDI), quantifying mean distance from the harm node, increases, indicating emergence of a stable social safety margin. Boxplots show distributions; all effects are significant ($***p < .001$, independent-samples t-test; 95% CI).

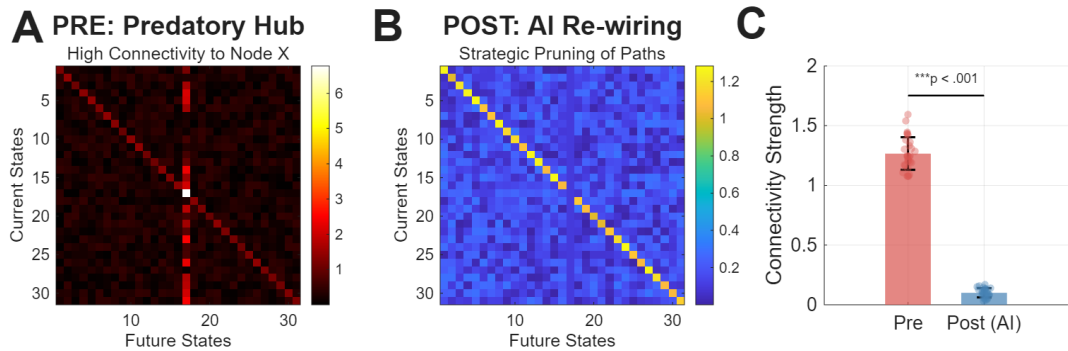


Figure 4: Structural Re-mapping of Predictive Neural Architecture via AI Intervention. (A) PRE: Predatory Neural Hub. The pre-intervention SR matrix heatmap reveals high predictive weights (hot colors) concentrated around the harmful node (X), driving the agent toward predatory interactions. (B) POST: AI-Scaffolded Re-wiring. Following AI-driven reward re-shaping, the SR matrix shows strategic pruning of node X. The resulting "cold" zone indicates a shift in connectivity toward prosocial navigation goals. (C) Neural Connectivity Pruning Effect. Quantitative comparison shows a significant reduction in mean connectivity strength to node X ($p < .001$, independent samples t-test). Scatter points represent variance across 30 simulated trials; error bars indicate SD. Heatmaps display states 40–70 for visual clarity.

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